

American International University-Bangladesh (AIUB)

Faculty of Science and Technology

Department of Computer Science

Web Technologies

CSE 3215

PROJECT REPORT

Project Title: “Online Medicine Shop”.

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Section: L

Submitted To:

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Group Information

Fill in all fields below. Each group member must sign before submission.

Group Number: 04

Project Title: Online Medicine Shop

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Declaration: We hereby declare that this project report is our original work. Any external resources or references used have been properly cited. We have not engaged in any form of academic dishonesty.

Marks Sheet

To be filled by the instructor only. Students must NOT write in this section.

Group No.	04	Section	L
Member 1 ID	23-55576-3		
Member 2 ID	23-55572-3		
Member 3 ID	23-52681-2		

CO3: Illustrate Multi-Tier Web Application for Targeted Society					
Feature Implementation (0-3)			Feature Implementation (0-3)		Completeness (0-3)
Basic Web Security	UI (HTML/CSS)	Feature Completeness	DB	Auth (Session/cookie)	MVC
CO4: Coordinate with Group Members to Develop Multi-Tier Application					
Technical Knowledge (0-3)			Collaborative Teamwork (0-3)		Communication/Promptness (0-3)
JS validation	PHP Validation	Ajax/JSON	Git Contribution		VIVA
Expected Outcome (0-2)		CO3 Total	CO4 Total		Project Total

Evaluated By: _____

Date : _____

Course Teacher Signature: _____

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Chapter 1: Project Overview

1.1 Project Description

Online Medicine Shop is a multi-tier web application developed for medicine browsing, ordering and administration. The system provides two user roles: admin and customer. Customers can register, login, browse medicines by category, search medicines, add items to cart, checkout with address, select payment method and view order status. Admin users can manage categories, medicines, customers, purchase requests and purchase history. The main purpose of the project is to reduce manual medicine ordering work by creating a simple online platform where products, orders and payments can be managed from one database-driven system.

Project Objectives

- Develop a role-based online medicine shop for admin and customer users.
- Implement registration, login, session management, profile update and logout.
- Display medicines with category-based browsing and AJAX medicine search.
- Allow the admin to manage medicine and category records using CRUD operations.
- Allow customers to add medicines to cart, update quantities, checkout and confirm orders.
- Use MySQL database tables with proper relationships among users, medicines, cart, orders and payments.
- Apply server-side validation, client-side validation, prepared statements and session-based access control.

1.2 Technologies Used

List all technologies, frameworks, and tools used in this project.

Layer	Technology / Tool	Purpose
Frontend	HTML5, CSS3	User interface structure and page design
Client-side	JavaScript / AJAX	Form validation, dynamic medicine search, cart actions and order status update
Backend	PHP	Controller logic, model functions and page routing
Database	MySQL / MariaDB	Persistent storage for users, medicines, cart, orders and payments
Database API	Procedural MySQLi with prepared statements	Secure database query execution
Authentication	PHP Sessions and Cookies	Login state, role checking and remember/search cookie support
Server	XAMPP / Apache	Local development and execution environment
Version Control	Git / GitHub	Team collaboration and contribution tracking

1.3 System Architecture / MVC Diagram

The project uses a basic MVC pattern. The front controller `index.php` receives page requests and loads config, models and controllers. Controllers validate user input and call model functions. Models communicate with the MySQL database. Views display HTML pages with CSS and JavaScript.

Name	Date modified	Type	Size
asset	5/19/2026 4:12 AM	File folder	
controllers	5/18/2026 7:38 PM	File folder	
css	5/18/2026 7:38 PM	File folder	
models	5/18/2026 7:38 PM	File folder	
uploads	5/18/2026 7:48 PM	File folder	
views	5/18/2026 7:38 PM	File folder	
config.php	5/19/2026 4:24 AM	PHP Source File	1 KB
database.sql	5/19/2026 12:59 AM	SQL File	5 KB
index.php	5/18/2026 7:38 PM	PHP Source File	3 KB
update_existing_database.sql	5/18/2026 7:38 PM	SQL File	1 KB

Figure 1: MVC architecture of the Online Medicine Shop system.

1.4 Database Schema

The application uses the database `medicine_shop_db`. The shared database includes users, categories, medicines, cart, orders, order_items and payments tables. The relationships allow one user to have many cart items and orders, one category to contain many medicines, one order to contain many order items, and one order to have one payment record.

Table Name	Important Fields	Description
users	id, name, email, password_hash, role, profile_picture, address, phone, remember_token, created_at	Stores admin and customer accounts, login credentials and profile details.
categories	id, name, category_type, image_path, created_at	Stores medicine categories and liquid/solid segmentation.
medicines	id, name, category_id, vendor_name, price, availability, description, image_path, created_at	Stores all product details shown to customers and managed by admin.
cart	id, user_id, medicine_id, quantity, added_at	Stores customer cart items before checkout.
orders	id, user_id, total_amount, shipping_address, status, payment_method, order_date	Stores customer order request and admin approval status.
order_items	id, order_id, medicine_id, quantity, unit_price	Stores individual medicine lines under an order.
payments	id, order_id, amount, payment_method, transaction_id, payment_date	Stores payment information and generated transaction ID.

Chapter 2: Git Contribution & Task Allocation

2.1 Task Allocation

Member	Student ID	Assigned Task	Implemented Module
MD. Rasel Hossain	23-55576-3	Task 1	Authentication, profile, home, category browsing and AJAX search
Mohammad Hasib Khandaker	23-55572-3	Task 2	Admin dashboard, category/medicine CRUD, customers, orders and history
Esrat Jahan Mukti	23-52681-2	Task 3	Cart, checkout, invoice, payment and customer orders

Task 1 main files: controllers/AuthController.php, controllers/HomeController.php, controllers/ProfileController.php, models/UserModel.php, models/MedicineModel.php, views/login.php, views/register.php, views/profile.php and views/home.php.

Task 2 main files: controllers/AdminController.php, models/CategoryModel.php, models/MedicineModel.php, models/OrderModel.php and views/admin/*.php.

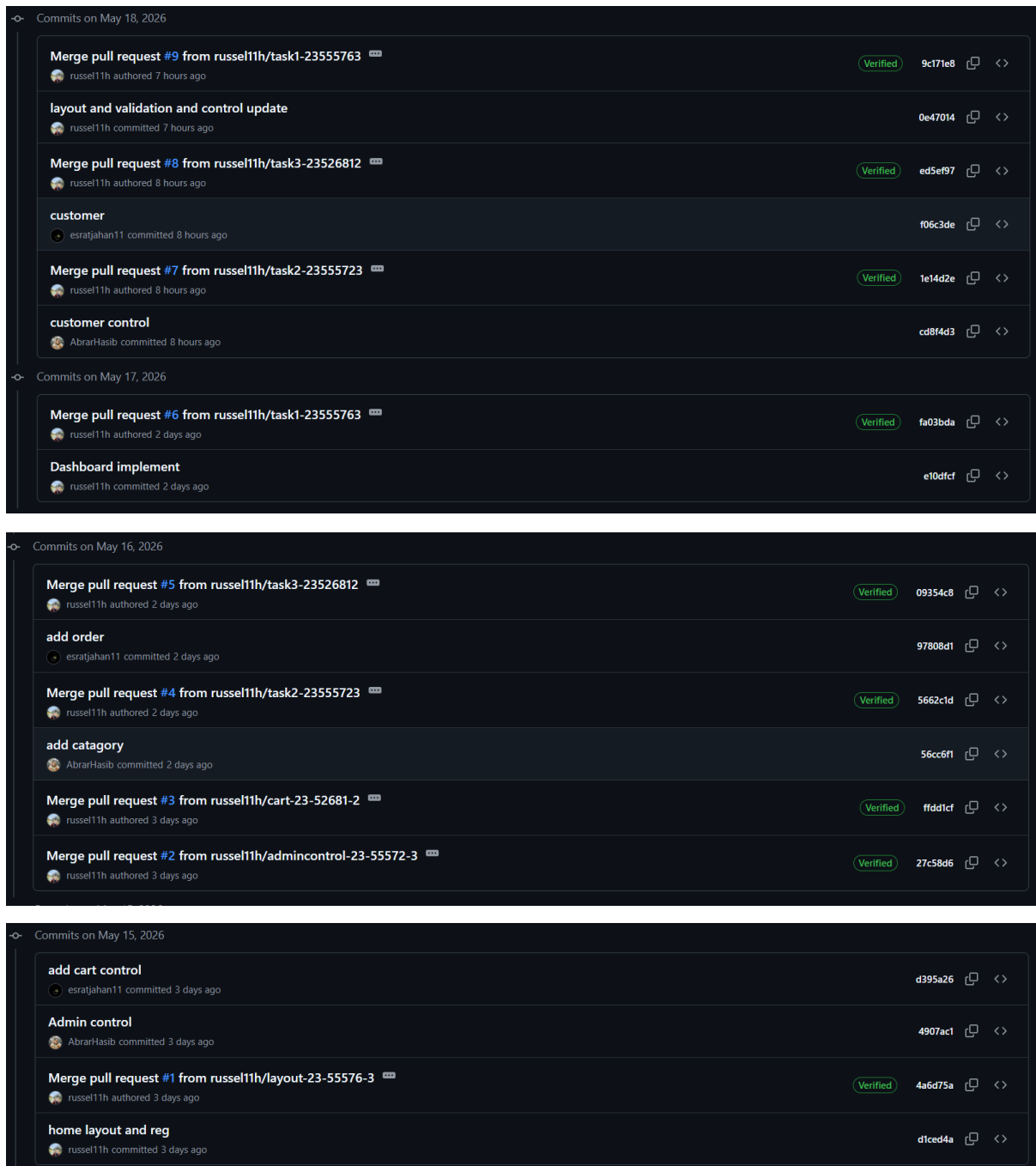
Task 3 main files: controllers/CartController.php, models/CartModel.php, models/OrderModel.php and views/customer/*.php.

2.2 GitHub Repository Link

Repository URL: <https://github.com/russell11h/wbt-spring-2025-2026.git>

2.3 Recommended Git Branch and Commit Records

Member	Branch Name	Minimum Commit Evidence
Rasel	task1-23555763 & layout-23-55576-3	home layout and reg, Dashboard implement, layout and validation and control update.
Hasib	task2-23555723 & admincontrol-23-55572-3	Admin control, add category, customer control.
Mukti	task3-23526812 & cart-23-52681-2	add cart control, add order, customer.



Screenshot instruction: Insert screenshot of GitHub commit history or contribution graph showing at least three meaningful commits from each member and merged branches into main.

Chapter 3: Features, Source Code and Screenshot Instructions

This chapter presents the main features implemented under Task 1, Task 2 and Task 3. Each feature includes the related source file path, key source code and explanation. Screenshot placeholders are kept so the final UI screenshots can be inserted after running the project in browser.

Feature 1: Registration, Login, Session and Logout

Task owner: Task 1 – Md. Rasel Hossain(23-55576-3)

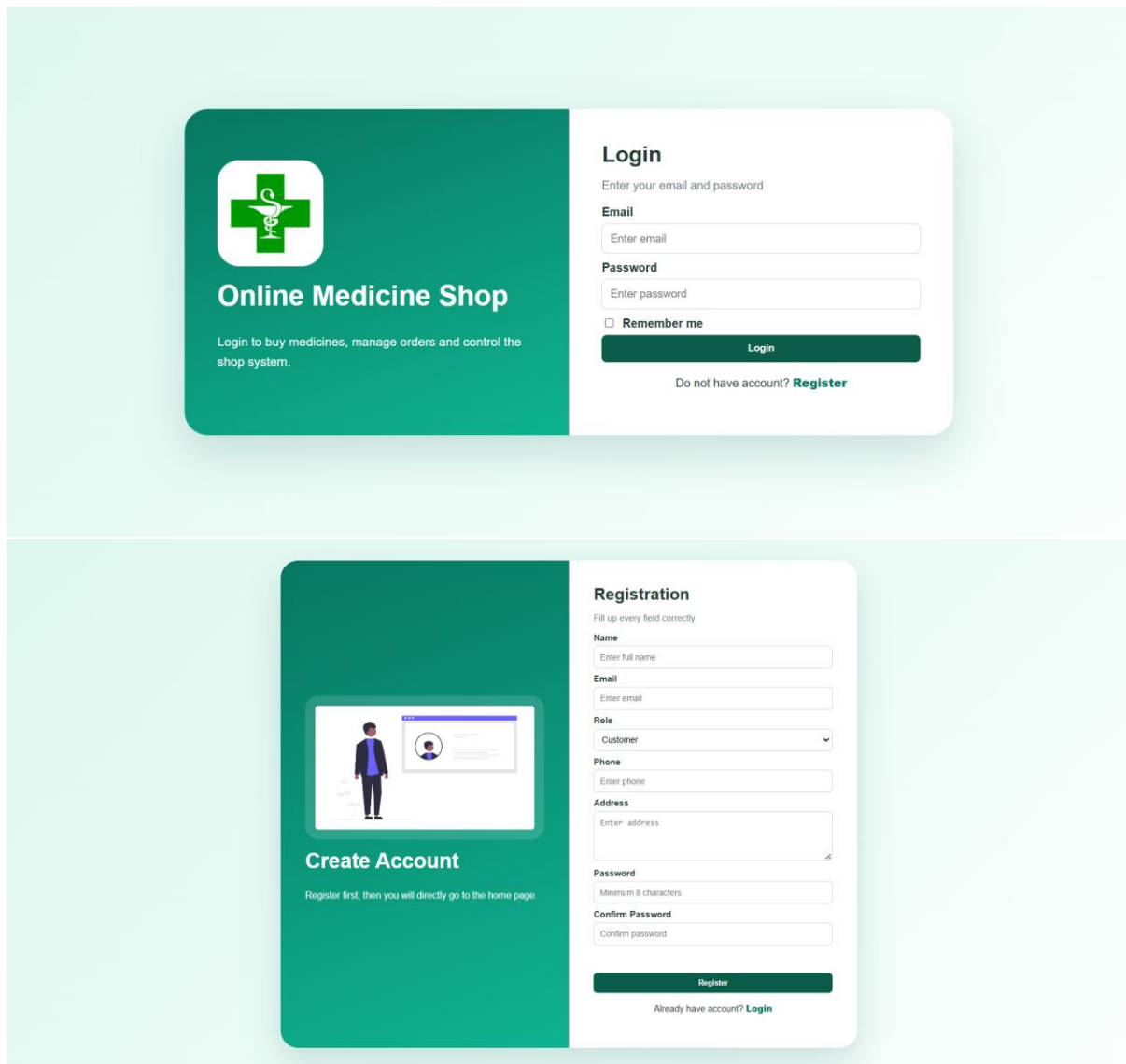
This feature allows both admin and customer users to create accounts and login. Registration validates required fields, email format, password length, confirm password and unique email. Login verifies the stored hash using password_verify() and stores the user identity in PHP session. Logout clears the session and remember-email cookie.

File: controllers/AuthController.php

```
function loginCtrl($conn) {
    $email = trim($_POST['email'] ?? '');
    $password = $_POST['password'] ?? '';
    $user = findUserByEmail($conn, $email);
    if ($user && password_verify($password, $user['password_hash'])) {
        $_SESSION['user'] = [
            'id' => $user['id'], 'name' => $user['name'],
            'email' => $user['email'], 'role' => $user['role']
        ];
        header('Location: index.php?page=home');
        exit;
    }
    $error = 'Invalid email or password.';
}

function registerCtrl($conn) {
    if (strlen($password) < 8) $error = 'Password must be at least 8 characters.';
    elseif ($password !== $confirm) $error = 'Passwords do not match.';
    elseif (emailExists($conn, $email)) $error = 'Email already exists.';
    else createUser($conn, $name, $email, $password, $role, $address, $phone);
}
```

Code Explanation: The controller reads user inputs, performs server-side validation and calls UserModel functions. password_verify() compares the submitted password with the hashed password in the database. On success, session values are created and role-based pages can identify the logged-in user.



Screenshot instruction: Insert two screenshots: one of the login page with email/password/remember me, and one of the registration page showing name, email, password, confirm password, address, phone and role dropdown.

Feature 2: Profile Management with Image Upload

Task owner: Task 1 - Md. Rasel Hossain (23-55576-3)

This feature allows logged-in admin and customer users to update profile information, upload a profile picture and change password after providing the current password. The profile picture upload validates file type and size before saving to uploads/profiles/.

File: controllers/ProfileController.php and models/UserModel.php

```
function uploadProfilePicture($fieldName, $oldPicture = "") {
    $allowedExt = ['jpg', 'jpeg', 'png', 'webp'];
    $ext = strtolower(pathinfo($_FILES[$fieldName]['name'], PATHINFO_EXTENSION));
    if (!in_array($ext, $allowedExt)) return ['ok'=>false, 'message'=>'Invalid image'];
    if ($_FILES[$fieldName]['size'] > 2 * 1024 * 1024) return ['ok'=>false,
'message'=>'Max 2MB'];
    $newPath = 'uploads/profiles/profile_' . time() . '_' . rand(1000,9999) . '.' . $ext;
```

```

move_uploaded_file($_FILES[$fieldName]['tmp_name'], $newPath);
return ['ok'=>true, 'path'=>$newPath];
}

function updateUserProfile($conn, $id, $name, $email, $address, $phone, $picture = null) {
    $stmt = mysqli_prepare($conn, "UPDATE users SET name=?, email=?, address=?,
phone=?, profile_picture=? WHERE id=?");
    mysqli_stmt_bind_param($stmt, 'sssssi', $name, $email, $address, $phone, $picture,
$id);
    return mysqli_stmt_execute($stmt);
}

```

Code Explanation: The upload function checks extension and file size before moving the image to the profile upload folder. The model uses a prepared UPDATE query so that profile data is saved safely for the logged-in user only.

The screenshot displays the 'My Profile' section of a web application. At the top, there's a navigation bar with 'Medicine Shop', a search bar, and links for 'My Orders', 'Rasel hossain', and 'Logout'. The left sidebar lists various medicine categories. The main content area is titled 'My Profile' and includes a sub-header 'Update your personal information, password and profile picture.' Below this, there's a profile card for 'Rasel hossain' with a placeholder image of an Android robot. To the right of the card are input fields for 'Name' (Rasel hossain), 'Email' (hossain.rasel2002@gmail.com), 'Phone' (01712345678), and 'Address' (488/1, Kuratoli). There's also a 'Profile Picture' section with a 'Choose file' button. Below the profile card is a 'Password Change' section with a note to fill in the fields only when changing the password. It contains 'Current Password' and 'New Password' (with a minimum 8 characters requirement) fields. An 'Update Profile' button is at the bottom.

Screenshot instruction: Insert screenshot of the profile page after login. The screenshot should show current profile image/logo, name, email, phone, address, image upload field and password change section.

Feature 3: Home Page, Category Browsing and AJAX Medicine Search

Task owner: Task 1 - Md. Rasel Hossain (23-55576-3)

The home page displays medicine cards and a category list. Users can click a category to filter medicines. The search box uses AJAX so medicine results update without refreshing the entire page. The last search text is stored in session and cookie.

File: controllers/HomeController.php and models/MedicineModel.php

```

function homeCtrl($conn) {
    $categories = getCategories($conn);
    $categoryId = intval($_GET['category'] ?? 0);
    $q = trim($_GET['q'] ?? ($_SESSION['last_search'] ?? ($_COOKIE['last_search'] ??
"")));
    $_SESSION['last_search'] = $q;
    setcookie('last_search', $q, time() + (86400 * 7), '/');
    $medicines = searchMedicinesByNameAndCategory($conn, $q, $categoryId);
}

```

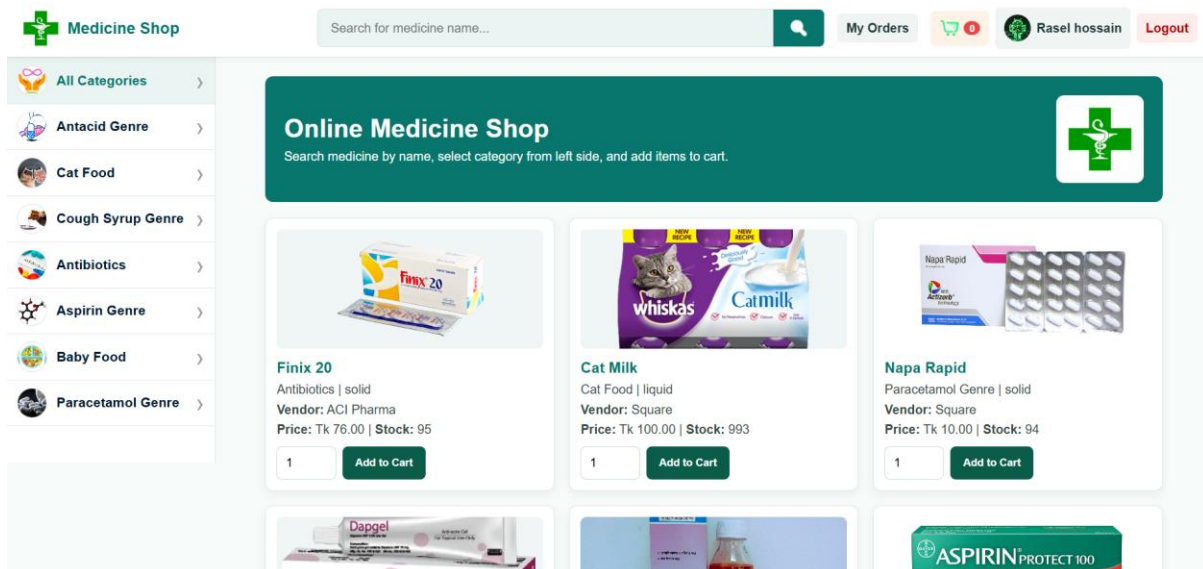
```

require 'views/home.php';
}

function medicineSearchAjax($conn) {
    header('Content-Type: application/json');
    $q = trim($_GET['q'] ?? '');
    $categoryId = intval($_GET['category'] ?? 0);
    echo json_encode(searchMedicinesByNameAndCategory($conn, $q, $categoryId));
    exit;
}

```

Code Explanation: The home controller loads categories and searches medicines by name and category. `medicineSearchAjax()` returns JSON. The JavaScript on the home view receives the JSON data and redraws the medicine cards dynamically.



Screenshot instruction: Insert screenshot of the home page showing left-side category buttons, top search box with search icon/button, cart icon, profile/logout buttons and medicine cards with images.

Feature 4: Admin Dashboard

Task owner: Task 2 - Mohammad Hasib Khandaker (23-55572-3)

The admin dashboard is the entry point for the admin module. It shows summary counts of medicines, categories, customers and pending orders. All admin pages are protected by `requireAdmin()`.

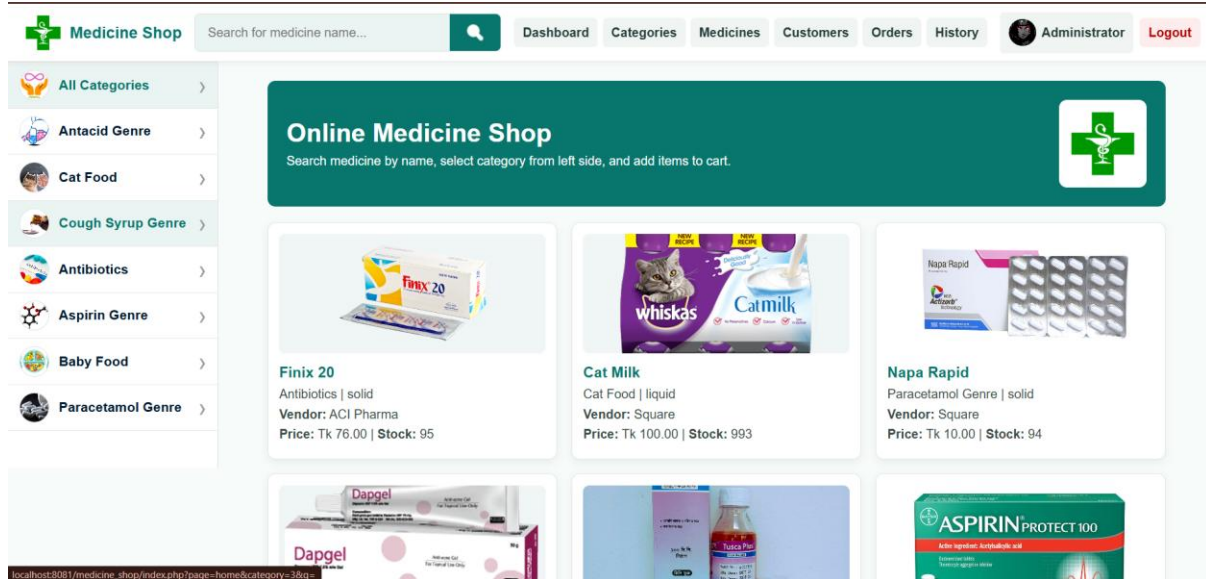
File: controllers/AdminController.php

```

function adminDashboardCtrl($conn) {
    requireAdmin();
    $stats = [
        'medicines' => countMedicines($conn),
        'categories' => countCategories($conn),
        'customers' => countCustomers($conn),
        'pending' => countPendingOrders($conn)
    ];
    require 'views/admin/dashboard.php';
}

```

Code Explanation: requireAdmin() blocks non-admin users. The dashboard collects total counts from model functions and passes the data to the admin dashboard view.



Screenshot instruction: Insert screenshot of admin dashboard showing four summary cards: total medicines, categories, customers and pending orders.

Feature 5: Category Management with Image and Type

Task owner: Task 2 - Mohammad Hasib Khandaker (23-55572-3)

The admin can add, edit and delete categories. Every category has a name, liquid/solid category type and category image. Deletion is blocked when medicines already exist under that category.

File: controllers/AdminController.php and models/CategoryModel.php

```
function categoryCtrl($conn) {
    requireAdmin();
    $action = $_GET['action'] ?? 'list';
    if ($action === 'add' && $_SERVER['REQUEST_METHOD'] === 'POST') {
        $name = trim($_POST['name'] ?? '');
        $type = $_POST['category_type'] ?? '';
        if ($name === '' || !in_array($type, ['liquid', 'solid'])) {
            $error = 'Valid category name and type are required.';
        } else {
            $upload = uploadCategoryImage('category_image');
            addCategory($conn, $name, $type, $upload['path']);
        }
    }
    $categories = getCategories($conn);
    require 'views/admin/categories.php';
}

function deleteCategory($conn, $id) {
    if (categoryHasMedicines($conn, $id)) return false;
}
```

```

$stmt = mysqli_prepare($conn, "DELETE FROM categories WHERE id = ?");
mysqli_stmt_bind_param($stmt, 'i', $id);
return mysqli_stmt_execute($stmt);
}

```

Code Explanation: The controller handles add, edit and delete requests. The model checks whether a category has medicines before deleting it, which protects database integrity.

Medicine Shop Search for medicine name... Dashboard Categories Medicines Customers Orders History Administrator Logout

All Categories

- Antacid Genre
- Cat Food
- Cough Syrup Genre
- Antibiotics
- Aspirin Genre
- Baby Food
- Paracetamol Genre

Manage Categories
Add or edit medicine categories with image, name and type.

Add Category

Category Name:

Type:

Category Image:

No	Image	Name	Type	Action
1		Antacid Genre	liquid	Edit Delete
2		Cat Food	liquid	Edit Delete

Screenshot instruction: Insert screenshot of category page showing category form with image upload, category type dropdown, category table, edit button and delete button.

Feature 6: Medicine Management CRUD

Task owner: Task 2 - Mohammad Hasib Khandaker (23-55572-3)

The admin can create, read, update and delete medicines. The medicine form includes medicine name, category, vendor, price, stock availability, description and image upload. Pending order checking prevents deletion of medicines used in active pending orders.

File: controllers/AdminController.php and models/MedicineModel.php

```

function addMedicine($conn, $name, $categoryId, $vendor, $price, $stock, $description, $image = null) {
    $stmt = mysqli_prepare($conn, "INSERT INTO medicines
        (name, category_id, vendor_name, price, availability, description, image_path)
        VALUES (?, ?, ?, ?, ?, ?, ?)");
    mysqli_stmt_bind_param($stmt, 'sisdiis', $name, $categoryId, $vendor, $price, $stock, $description, $image);
    return mysqli_stmt_execute($stmt);
}

function deleteMedicine($conn, $id) {
    if (medicineInPendingOrder($conn, $id)) return false;
    $stmt = mysqli_prepare($conn, "DELETE FROM medicines WHERE id = ?");
    mysqli_stmt_bind_param($stmt, 'i', $id);
    return mysqli_stmt_execute($stmt);
}

```

Code Explanation: The add function inserts a new medicine using a prepared statement. The delete function first checks pending order usage and then removes the record safely.

The screenshot displays the 'Medicine Shop' interface. On the left is a sidebar with category links: All Categories, Antacid Genre, Cat Food, Cough Syrup Genre, Antibiotics, Aspirin Genre, Baby Food, and Paracetamol Genre. The main area is titled 'Manage Medicines' with a subtitle 'Add medicine name, category, vendor, price, stock and image.' Below this is an 'Add Medicine' form with fields for Name, Category (dropdown), Vendor, Price, Stock, Medicine Image (file upload), and Description. An 'Add Medicine' button is at the bottom of the form.

Below the form is a table listing existing medicines. The table has columns: No, Image, Name, Category, Vendor, Unit Price, Stock, and Action. The Action column contains 'Edit' and 'Delete' buttons for each medicine.

No	Image	Name	Category	Vendor	Unit Price	Stock	Action
1		Finix 20	Antibiotics	ACI Pharma	Tk 76.00	95	Edit Delete
2		Cat Milk	Cat Food	Square	Tk 100.00	993	Edit Delete
3		Napa Rapid	Paracetamol Genre	Square	Tk 10.00	94	Edit Delete
4		DP Gel Suspension	Antacid Genre	ACI Pharma	Tk 120.00	0	Edit Delete
5		Tusca Plus Syrup	Cough Syrup Genre	Square Pharma	Tk 85.00	50	Edit Delete
6		Aspirin Protect	Aspirin Genre	Bayer	Tk 5.00	90	Edit Delete
7		Ace Plus	Paracetamol Genre	Square Pharma	Tk 3.50	149	Edit Delete

Screenshot instruction: Insert screenshot of medicine management page showing add/edit form, category dropdown, medicine image upload, medicine table, edit button and delete button.

Feature 7: Customer List and Delete Customer

Task owner: Task 2 - Hasib (23-55572-3)

The admin can view registered customers and delete a customer account. Customer deletion uses database cascade relations for cart and orders connected with the customer.

File: controllers/AdminController.php and models/UserModel.php

```
function customersCtrl($conn) {
    requireAdmin();
    if (($GET['action'] ?? '') === 'delete') {
        deleteCustomer($conn, intval($GET['id'] ?? 0));
        header('Location: index.php?page=customers&msg=deleted');
        exit;
    }
}
```

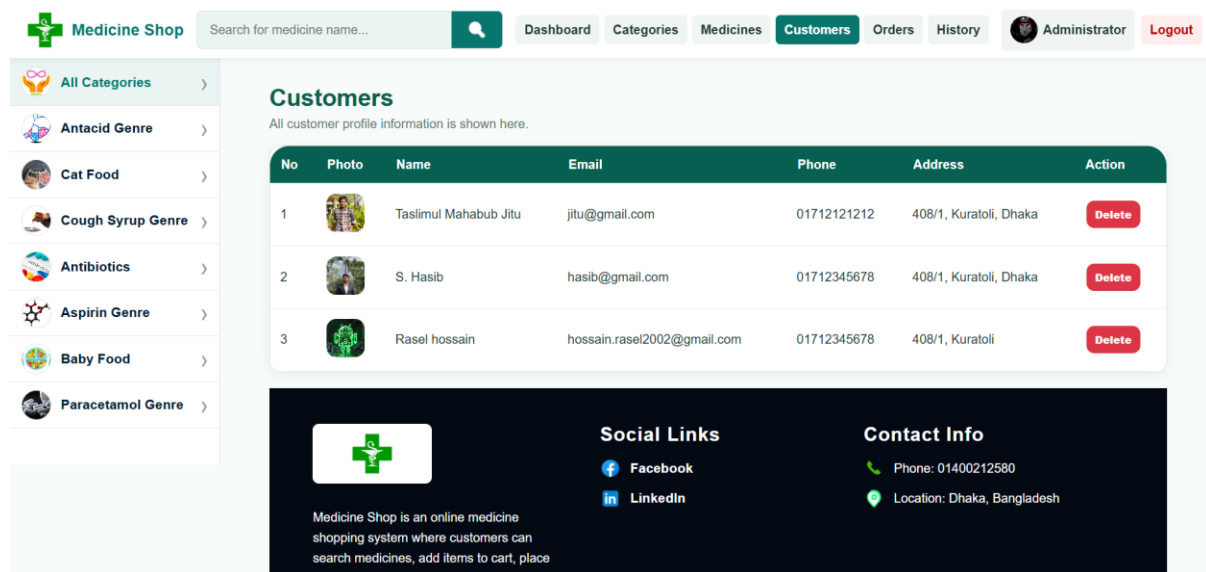
```

    }
    $customers = getAllCustomers($conn);
    require 'views/admin/customers.php';
}

function getAllCustomers($conn) {
    $result = mysqli_query($conn, "SELECT * FROM users WHERE role='customer'
    ORDER BY id DESC");
    return mysqli_fetch_all($result, MYSQLI_ASSOC);
}

```

Code Explanation: The controller verifies admin access, processes delete actions and loads all customer records. The customer model only selects users where role is customer.



Screenshot instruction: Insert screenshot of admin customer list with customer name, email, phone, address and delete button.

Feature 8: Purchase Requests and AJAX Accept/Reject

Task owner: Task 2 - Mohammad Hasib Khandaker (23-55572-3)

The admin can see all purchase requests with customer details and order items. Order status can be changed to accepted or rejected using AJAX without reloading the page.

File: controllers/AdminController.php and models/OrderModel.php

```

function orderStatusAjax($conn) {
    requireAdmin();
    header('Content-Type: application/json');
    $orderId = intval($_POST['order_id'] ?? 0);
    $status = $_POST['status'] ?? '';
    if (!in_array($status, ['accepted', 'rejected'])) {
        http_response_code(422);
        echo json_encode(['ok' => false]);
        exit;
    }
    updateOrderStatus($conn, $orderId, $status);
    echo json_encode(['ok' => true, 'status' => $status]);
}

```



```

exit;
}

function updateOrderStatus($conn, $orderId, $status) {
    $stmt = mysqli_prepare($conn, "UPDATE orders SET status = ? WHERE id = ?");
    mysqli_stmt_bind_param($stmt, 'si', $status, $orderId);
    return mysqli_stmt_execute($stmt);
}

```

Code Explanation: The AJAX controller returns JSON. It validates that only accepted or rejected status values are allowed, then calls the model to update the database.

The screenshot shows the 'Medicine Shop' admin dashboard. On the left is a sidebar with categories like 'All Categories', 'Antacid Genre', 'Cat Food', 'Cough Syrup Genre', 'Antibiotics', 'Aspirin Genre', 'Baby Food', and 'Paracetamol Genre'. The main content area is titled 'Purchase Requests' and contains a table of customer orders. The table has columns: ID, Customer, Total, Transaction ID, Address, Date, Status, and Action. The status column shows 'rejected', 'pending', and 'accepted' with corresponding buttons to 'Accept' or 'Reject'.

ID	Customer	Total	Transaction ID	Address	Date	Status	Action
#13	Rasel hossain hossain.rasel2002@gmail.com	Tk 152.00	TXN-1779129986-1891	408/1, Kuratoli	2026-05-19 00:46:25	rejected	Done
#12	Rasel hossain hossain.rasel2002@gmail.com	Tk 107.00	TXN-1779127650-2997	408/1, Kuratoli, Dhaka	2026-05-19 00:07:30	pending	Accept Reject
#11	Rasel hossain hossain.rasel2002@gmail.com	Tk 3,600.00	TXN-1779124994-3209	408/1, Kuratoli	2026-05-18 23:23:14	accepted	Done
#10	Rasel hossain hossain.rasel2002@gmail.com	Tk 152.00	TXN-1779124924-9336	408/1, Kuratoli	2026-05-18 23:22:04	pending	Accept Reject
#9	Rasel hossain hossain.rasel2002@gmail.com	Tk 76.00	TXN-1779123687-8026	408/1, Kuratoli	2026-05-18 23:01:27	pending	Accept Reject
#8	Rasel hossain hossain.rasel2002@gmail.com	Tk 120.00	TXN-1779122133-6981	408/1, Kuratoli, Dhaka	2026-05-18 22:35:33	rejected	Done
#7	Rasel hossain	Tk 400.00	TXN-1779104736-	408/1, Kuratoli	2026-05-18	pending	Accept Reject

Screenshot instruction: Insert screenshot of admin orders page showing customer name, total amount, shipping address, medicine list, current status, accept button and reject button.

Feature 9: All Customers Purchase History

Task owner: Task 2 - Mohammad Hasib Khandaker (23-55572-3)

This feature shows completed orders that have been accepted by admin. It helps the admin review customer purchases, order items, quantities, unit price, total price and payment transaction ID.

File: controllers/AdminController.php and models/OrderModel.php

```

function historyCtrl($conn) {
    requireAdmin();
    $orders = attachOrderDetails($conn, getAcceptedOrders($conn));
    require 'views/admin/history.php';
}

function getAcceptedOrders($conn) {
    $sql = "SELECT o.*, u.name AS customer_name, u.email, u.phone
    FROM orders o JOIN users u ON o.user_id = u.id
    WHERE o.status = 'accepted' ORDER BY o.id DESC";
}

```



```

$result = mysqli_query($conn, $sql);
return mysqli_fetch_all($result, MYSQLI_ASSOC);
}

```

Code Explanation: getAcceptedOrders() loads only accepted orders. attachOrderDetails() adds medicine items and payment details before passing the list to the history view.

Medicine Shop Search for medicine name... Dashboard Categories Medicines Customers Orders **History** Administrator Logout

Admin Order History
Only accepted orders are shown here.

Order #11 - Rasel hossain
Total: Tk 3,600.00 | Transaction ID: TXN-1779124994-3209 | Date: 2026-05-18 23:23:14

Medicine	Qty	Unit Price	Total Price
DP Gel Suspension	30	Tk 120.00	Tk 3,600.00

Order #6 - Rasel hossain
Total: Tk 100.00 | Transaction ID: TXN-1779104736-6938 | Date: 2026-05-18 17:45:36

Medicine	Qty	Unit Price	Total Price
Cat Milk	1	Tk 100.00	Tk 100.00

Order #5 - Rasel hossain

Screenshot instruction: Insert screenshot of accepted purchase history showing customer details, medicine list, quantities, unit price, total amount and transaction ID.

Feature 10: Add to Cart and Cart Management

Task owner: Task 3 - Esrat Jahan Mukti (23-52681-2)

Customers can add medicines to the cart from product cards, increase/decrease quantity and remove items. AJAX endpoints return JSON to update the cart count and page totals.

File: controllers/CartController.php and models/CartModel.php

```

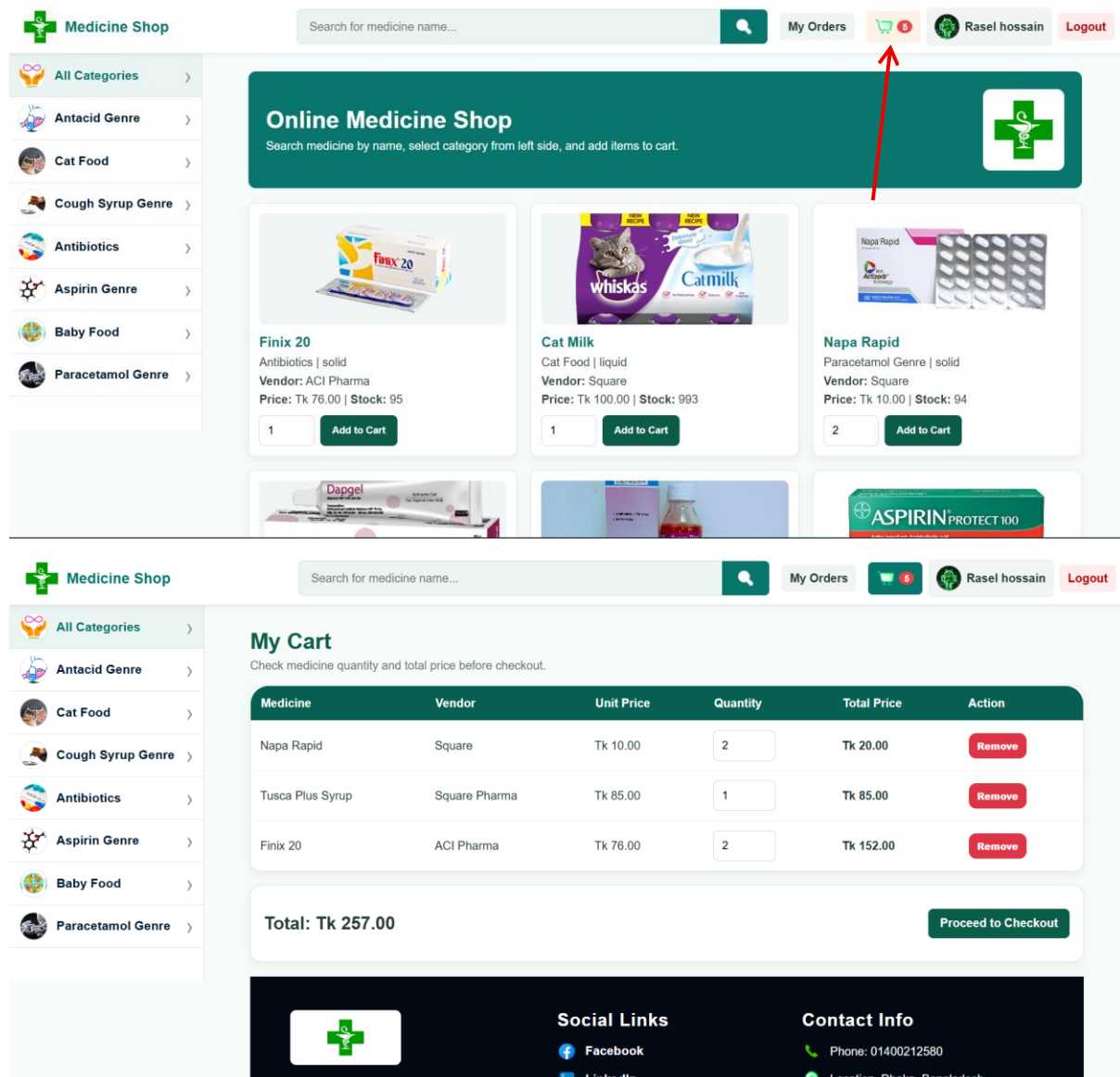
function cartAddAjax($conn) {
    requireCustomer();
    header('Content-Type: application/json');
    $medicineId = intval($_POST['medicine_id'] ?? 0);
    $qty = intval($_POST['quantity'] ?? 1);
    $medicine = getMedicine($conn, $medicineId);
    if (!$medicine || $qty <= 0 || $qty > intval($medicine['availability'])) {
        http_response_code(422);
        echo json_encode(['ok'=>false, 'message'=>'Invalid quantity or medicine stock.']);
        exit;
    }
    addToCart($conn, $_SESSION['user']['id'], $medicineId, $qty);
    echo json_encode(['ok'=>true, 'count'=>cartCount($conn, $_SESSION['user']['id'])]);
    exit;
}

function addToCart($conn, $userId, $medicineId, $qty) {
    $stmt = mysqli_prepare($conn, "INSERT INTO cart (user_id, medicine_id, quantity)

```

```
VALUES (?, ?, ?) ON DUPLICATE KEY UPDATE quantity = quantity +
VALUES(quantity)");
mysqli_stmt_bind_param($stmt, 'iii', $userId, $medicineId, $qty);
return mysqli_stmt_execute($stmt);
}
```

Code Explanation: The controller checks customer login, validates medicine existence and stock, then updates the cart. The model uses ON DUPLICATE KEY UPDATE so the same medicine increases quantity instead of creating duplicate rows.



Screenshot instruction: Insert screenshot of medicine card with add-to-cart button and screenshot of cart page showing item name, vendor, unit price, quantity buttons, remove option, subtotal and grand total.

Feature 11: Checkout, Invoice and Payment Method

Task owner: Task 3 - Esrat Jahan Mukti (23-52681-2)

The checkout page collects shipping address, then the invoice page shows medicine details and total amount. Customer must select payment method before order confirmation.

File: controllers/CartController.php and models/OrderModel.php

```

function checkoutCtrl($conn) {
    requireCustomer();
    $items = getCartItems($conn, $_SESSION['user']['id']);
    if ($_SERVER['REQUEST_METHOD'] === 'POST') {
        $address = trim($_POST['shipping_address'] ?? '');
        if ($address === '') $error = 'Shipping address is required.';
        else {
            $_SESSION['checkout_address'] = $address;
            header('Location: index.php?page=invoice');
            exit;
        }
    }
    require 'views/customer/checkout.php';
}

function createOrder($conn, $userId, $total, $address, $paymentMethod) {
    $stmt = mysqli_prepare($conn, "INSERT INTO orders
        (user_id, total_amount, shipping_address, payment_method, status)
        VALUES (?, ?, ?, ?, 'pending')");
    mysqli_stmt_bind_param($stmt, 'idss', $userId, $total, $address, $paymentMethod);
    mysqli_stmt_execute($stmt);
    return mysqli_insert_id($conn);
}

```

Code Explanation: The checkout controller stores the address in session and redirects to invoice. createOrder() stores a pending order so the admin can later accept or reject it.

Medicine Shop

Search for medicine name...

My Orders Rassel hossain [Logout](#)

Checkout
Confirm your shipping address.

Shipping Address
408/1, Kuratoli, Dhaka

Total: Tk 257.00
[Show Invoice](#)

Social Links
[Facebook](#)
[LinkedIn](#)

Contact Info
 Phone: 01400212580
 Location: Dhaka, Bangladesh

Medicine Shop is an online medicine shopping system where customers can search medicines, add items to cart, place orders, and manage their profile easily.

Medicine Shop Search for medicine name... My Orders Rassel hassain Logout

Invoice & Payment
Review order and select one payment method.

Shipping Address: 408/1, Kuratoli, Dhaka

Medicine	Quantity	Unit Price	Total Price
Napa Rapid	2	Tk 10.00	Tk 20.00
Tusca Plus Syrup	1	Tk 85.00	Tk 85.00
Finix 20	2	Tk 76.00	Tk 152.00

Grand Total: Tk 257.00

Payment Method

☐ Credit Card
 ☐ bKash
 ☐ Nagad
 ☐ Bank Transfer
 ☐ Cash on Delivery

Cancel Confirm Purchase

Screenshot instruction: Insert screenshot of checkout address form and invoice/payment page showing ordered medicines, quantity, unit price, total amount, payment method radio buttons and confirm purchase button.

Feature 12: Order Confirmation and My Orders

Task owner: Task 3 - Esrat Jahan Mukti (23-52681-2)

After successful payment selection, the system creates order, order_items and payment record, reduces medicine stock, clears the cart and displays the customer order list with pending/accepted/rejected status.

File: controllers/CartController.php

```

mysqli_begin_transaction($conn);
try {
    $orderId = createOrder($conn, $_SESSION['user']['id'], $total, $address, $method);
    foreach ($items as $item) {
        addOrderItem($conn, $orderId, $item['id'], $item['quantity'], $item['price']);
        reduceMedicineStock($conn, $item['id'], $item['quantity']);
    }
    addPayment($conn, $orderId, $total, $method);
    clearCart($conn, $_SESSION['user']['id']);
    mysqli_commit($conn);
    header('Location: index.php?page=my_orders&success=1');
} catch (Exception $e) {
    mysqli_rollback($conn);
    $error = $e->getMessage();
}

```

Code Explanation: The transaction block keeps order creation reliable. If any insert or stock update fails, `mysqli_rollback()` cancels all changes. If everything works, `commit` saves the order and redirects the customer to My Orders.

Medicine Shop

Search for medicine name...

My Orders
Your purchase history and transaction information.

Purchase confirmed. Order is pending admin approval.

Order #14 - Pending
Order Total: Tk 257.00 | Payment: bKash | Transaction ID: TXN-1779150451-2743 | Date: 2026-05-19 06:27:31

Medicine	Quantity	Unit Price	Total Price
Napa Rapid	2	Tk 10.00	Tk 20.00
Tusca Plus Syrup	1	Tk 85.00	Tk 85.00
Finix 20	2	Tk 76.00	Tk 152.00

Order #13 - Rejected
Order Total: Tk 152.00 | Payment: Credit Card | Transaction ID: TXN-1779129986-1891 | Date: 2026-05-19 00:46:25

Medicine	Quantity	Unit Price	Total Price
----------	----------	------------	-------------

Screenshot instruction: Medicine list, total price, transaction ID, payment method and status pending admin approval or accepted/rejected.

Chapter 4: Conclusion & Future Work

4.1 Summary

The Online Medicine Shop project successfully implements a role-based multi-tier web application using PHP, MySQL, HTML, CSS and JavaScript. Task 1 completed user authentication, profile management, home page, category browsing and AJAX search. Task 2 completed admin features including dashboard, category management, medicine CRUD, customer management, purchase request handling and purchase history. Task 3 completed customer cart management, checkout, invoice, payment method selection and order confirmation. The system follows MVC separation and stores all important data in a relational database.

4.2 Challenges Faced

- Image upload path issues were solved by using consistent upload folders for profiles, categories and medicines.
- Role-based page access was handled by adding `requireAdmin()` and `requireCustomer()` checks before protected controller functions.
- AJAX operations were separated into JSON endpoints so search, cart and order status updates could work without full page refresh.
- Database relationships were maintained using foreign keys and cascade deletion where appropriate.
- UI consistency was improved by placing common header, footer and CSS files in shared layout and css folders.

4.3 Future Improvements

- Add medicine prescription upload and pharmacist verification for restricted medicines.
- Add online payment gateway integration for bKash/Nagad/card payment verification.
- Add email or SMS notification for order confirmation and admin approval status.
- Add advanced filter options such as price range, vendor, availability and category type.
- Add pagination and reporting dashboard for large medicine and order records.
- Add stronger CSRF token protection for every POST form and AJAX request.

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